

Calibration targets: what is verified?

September 22, 2026 | Target validation before further calibration

The frozen 13-row target/weight system was used consistently. Saved empirical receipts reproduce the measured values; 2.1 is an author normalization and 0.88% is an inherited external restriction. This is not a fresh raw-data replication.

The data-model mapping is not fully closed. Prioritize population/geography, the first-birth rooms estimand and its model counterpart, resident-child groups, and wealth/income concepts. Existing room caps and fractional age masks work in the actual scored code.

No targets or weights have been changed and no overnight utility-comparison jobs have been submitted. A longer fit should follow an explicitly frozen measurement contract.

Complete current target list

Moment	Current target	Remaining decision
Initial fertility normalization	2.1	Author normalization
Childless women, ages 40-44	19.8279%	Population/age approximation
Exactly one child among mothers	21.3655%	Population/age approximation
Mean first-birth age	25.9763 years	Period/boundary approximation
First births at age 30 or above	24.9278%	Period/population approximation
Aggregate wealth / annual earnings	6.14586	Components and dates
Annual bequests / aggregate wealth	0.88%	External source/precision
Old wealth/income p90 / median	3.51594	Income denominator mismatch
Mean occupied rooms, cap 9	5.56110	Geography/sample choice
Ownership, heads 30-55	64.8334%	Geography/structure sample
First-birth room response	0.720246 rooms	Estimator and model mapping
Rooms: 3+ versus 1-2 resident children	0.347067 rooms	Resident/dependent child proxy
Recent-parent ownership gap	16.2896 pp	Recent-child group proxy

Housing: source and model measurement

Mean occupied rooms, cap 9 | 5.56110

Data: ACS 2005/2006, selected 42 MET 2013 metros, household heads 18-85 and the saved housing-validity filters. Weighted rooms capped at 9. Reproduced exactly.

Model: Current realized owner/renter rooms capped at 9 in every income-resolved state before aggregation. Fractional age exposure is implemented.

Assessment: Cap/order implementation is verified. Decide national versus 42 metros and define the represented housing sample. Saved national counterpart is 5.607886.

Ownership, heads 30-55 | 64.8334%

Data: ACS 2005/2006 heads 30-55 in 42 metros, positive household weights and the specified occupied-unit/structure sample. Weighted ownership share; reproduced exactly.

Model: Current realized owner mass over total mass, with half of the 54-57 cell included. No whole-cell age-mask error.

Assessment: Within-cell age uniformity is explicit. Geography and ACS structure types need a model interpretation. Saved national counterpart is 67.6260%; switching geography does not cure low ownership.

First-birth room response | 0.720246 rooms

Data: Weighted national PSID household-aligned event study: coefficient at +3 minus coefficient at -1, omitted -2; person/year effects, age/education controls and person clustering. Stored scalar and SE reproduce.

Model: Matched treated/control branches advanced one four-year period under stationary policies, with destination continuation births allowed; uncapped rooms. This implemented calculation is not the same regression/cohort weighting.

Assessment: Not a settled target. Reference support and demonstrated normalization sensitivity remain; author preferred a -2 empirical baseline. Completed binned diagnostic 0.426502 is a different sample/design and is not a replacement. Fix the empirical estimand and model counterpart together.

Rooms: 3+ versus 1-2 resident children | 0.347067 rooms

Data: ACS heads 30-55 with youngest resident own child under 18; compare all resident-own-child counts 3+ versus 1-2, with rooms capped 9. Exact saved reproduction.

Model: Model current dependents 3+ versus 1-2, restricted to feasible count states; caps full-state rooms before aggregation.

Assessment: Resident own children can include adult children; current dependents with stochastic departure are not that exact grouping. Choose and justify the shared measurement.

Recent-parent ownership gap | 16.2896 pp

Data: ACS heads 30-55: ownership where oldest resident own child is under 4 minus ownership with no resident own children. Exact saved reproduction.

Model: Synchronized current births from previously empty-dependent homes minus current empty homes, including former parents. This is the active repaired observer.

Assessment: Not the obsolete lifetime-childless comparison. It remains an approximation to an interview stock defined by oldest child age. Sampling uncertainty alone does not measure this discrepancy. National counterpart 12.7608 pp is available.

Fertility: source and model measurement

Initial fertility normalization | 2.1

Data: Author-imposed stationary model normalization, not an empirical estimate.

Model: The existing normalization solves for the fertility utility scale; no weight in the twelve-row objective.

Assessment: Keep its role explicit. Do not interpret 2.1 as a directly measured period birth rate or equate it with every cohort fertility measure.

Childless women, ages 40-44 | 19.8279%

Data: June CPS 2004/2006, women 40-44, valid children-ever-born and positive fertility-supplement weight; weighted share with zero children. Saved source estimate reproduced.

Model: Uniform birth-time interpolation across four-year age cells, with fractional overlap at 40-44; uses the model household reproductive-member mass.

Assessment: Age mapper is implemented. Female population and empirical/model age composition need an explicit common interpretation.

Exactly one child among mothers | 21.3655%

Data: Same CPS sample; weighted share with exactly one child conditional on ever having a child.

Model: Exactly-one-child stock divided by the stock of mothers, using the same age projection.

Assessment: This is not a second-birth hazard. Reconcile population mapping alongside childlessness; no unavailable-mapper claim.

Mean first-birth age | 25.9763 years

Data: NCHS first-birth counts 2003-2006. Ages are mapped to the model midpoint classes; tails below 18 and above 45 are absorbed into boundary groups. Saved estimate reproduced.

Model: First-birth flow weighted mean of model age-cell midpoints in the stationary economy.

Assessment: Same grouped-age calculation is available. Stationary versus dated period, population/residence scope and boundary support remain explicit assumptions.

First births at age 30 or above | 24.9278%

Data: Share of pooled first births at exact maternal age 30 or above, among the saved age 12-49 counts.

Model: Share of model first-birth flow in cells beginning at 30 or above.

Assessment: Threshold matches a model cell boundary. Period/population and support assumptions still require a declared target interpretation.

Wealth: source and model measurement

Aggregate wealth / annual earnings | 6.14586

Data: National PSID 2003/2005: weighted total net worth of reference-person families 18-85 divided by weighted gross labor earnings for working-age families. Ratio of sums, not mean household ratio.

Model: Beginning net financial wealth plus owner housing value, divided by annualized gross model labor earnings. Four-year flows are annualized once and payroll tax is grossed up.

Assessment: Arithmetic and annualization are verified. Reconcile mortgage/net-worth components and stock versus income reference dates before treating this as an exact empirical counterpart.

Annual bequests / aggregate wealth | 0.88%

Data: Inherited external historical restriction. No in-repository microdata builder or empirical SE; this review verifies its stored value, not its primary-source derivation.

Model: Expected nonnegative estate flows at death, annualized once, divided by aggregate beginning wealth; terminal death is imposed.

Assessment: Verify the source definition and accept or revise the external restriction explicitly. Its 5% tolerance is synthetic, not estimated precision. No automatic dropping: preserve bequest-block identification.

Old wealth/income p90 / median | 3.51594

Data: National PSID 2003/2005 reference persons 76-84, family income above \$1000 and observed child history. Weighted 90th percentile of net-worth/family-income divided by its weighted median.

Model: Same quantile operation and explicit 76-84 cell overlap, but income is modeled pension plus configured transfer. The scorer unambiguously uses this observer, not the legacy label-age statistic.

Assessment: Family income contains components absent from the pension proxy. Income threshold, observed-history selection and wealth components need reconciliation. This is a substantive mapping issue, not an ambiguous active scorer.

Weights and decisions before launch

Current objective: sum of twelve squared gaps divided by fixed working scales squared. The separate 2.1 normalization is unscored. These scales are not all sampling standard errors; cross-moment covariance is not used.

Moment	Working scale	Basis
Initial fertility normalization	-	Unscored
Childless women, ages 40-44	0.00530504	Approximate pooled CPS uncertainty
Exactly one child among mothers	0.00609113	Approximate pooled CPS uncertainty
Mean first-birth age	0.0845674	Annual 2003-2006 temporal SD
First births at age 30 or above	0.00849226	Annual 2003-2006 temporal SD
Aggregate wealth / annual earnings	0.362855	PSID person bootstrap SE
Annual bequests / aggregate wealth	0.00044	Synthetic 5% of target
Old wealth/income p90 / median	0.306911	PSID person bootstrap SE
Mean occupied rooms, cap 9	0.0883812	42-metro bootstrap SE
Ownership, heads 30-55	0.0206753	42-metro bootstrap SE
First-birth room response	0.0852601	Clustered contrast SE
Rooms: 3+ versus 1-2 resident children	0.0597052	42-metro bootstrap SE
Recent-parent ownership gap	0.00607952	42-metro bootstrap SE

Recommended closure order

1. Choose the quantitative population: national US or the maintained 42 metro sample. National cross-sectional housing targets already exist. Decision pending author response.
2. Settle first-birth rooms: retain the intended empirical reference, handle observed support, and measure a comparable object in the model. Do not silently substitute a binned diagnostic or keep 0.720246 as a certified fact.
3. Resolve child-group and wealth-income definitions; distinguish acceptable declared model approximations from required empirical remeasurement. Preserve identification if any row is changed.
4. Document working weights by purpose, freeze the complete source/target/observer fingerprint, verify the exact measurement loop, then reconsider the two-utility overnight search.

Evidence and limits of this review

Actual scored inputs and observer

output/model/native_financing_diagnostic_20260919/specification_followup/earnings_entry_battery_v1/final_readout/B/selected/: target_fit.csv, native_summary.json, run_contract.json.

Fertility source verification

output/model/e5f_matched_pf_20260909a/design_research/fertility_contract/: fertility_target_contract.json and verification.json; saved CPS partitions and NCHS count tables.

Housing scalar reproduction

output/model/native_financing_diagnostic_20260919/specification_followup/housing_profiles_v1/full/target_recomputed.json and final_receipt.json.

PSID event-study measurement and reference work

code/data/psid_followup_mar2026/output/first_birth_correction_review/README.md, including completed all-wave, annual and binned comparisons; original 0.720246 target_receipt.csv in sa_rooms_first_birth_household_aligned_v1/.

Wealth builders and outputs

output/model/e5f_matched_pf_20260909a/design_research/wealth/: build_initial_wealth.R, aggregate_wealth_results.csv, old_wealth_results.csv and verification.json.

Weights and exact model code

lead_methodological_decision.json and frozen working_contract.json; frozen source code/model/tools/e5f_initial_housing_observer.py, e5f_initial_fertility_observer.py and scorer/score_initial.py.

Independent checks and lead corrections

This packet: fertility_receipt.json, housing_receipt.json, wealth_receipt.json, lead_integrity_review.json. Lead removed obsolete missing-cap/age-mask warnings and distinguished the active observer from legacy readouts.

This review reads saved empirical receipts and the actual scored observer/source. It does not rerun microdata, change estimates or weights, certify structural identification, or submit a model job. Detailed per-row receipts and the complete target catalogue accompany this PDF.